Zenith Structural Holdings LLC
3096 Wilson Drive
Petoskey, MI 49770

January 9, 2026

DARPA Tactical Technology Office (TTO) Attn: ALIAS-Texas Program Manager

675 North Randolph Street

Arlington, VA 22203

RE: Proposal for HR0011SB20254XL-01 – Agnostic Equalization Mechanism (AEM)

To the DARPA Selection Committee,

Zenith Structural Holdings LLC is pleased to submit this Direct-to-Phase II (DP2) proposal for the development and integration of the **Agnostic Equalization Mechanism (AEM)**. Our proposed solution addresses the critical safety gap in missionized autonomy by providing a hardware-rooted "Structural Veto" that protects the **ALIAS/MATRIX** core from the non-deterministic behaviors of third-party mission applications.

As demonstrated in the attached feasibility documentation, Zenith has already surpassed the technical merit typically associated with a Phase I effort. Through Hardware-in-the-Loop (HIL) testing conducted on **January 5, 2026**, we verified a sub-millisecond interdiction latency of **54.4μs**—well within the stringent stability requirements of the MATRIX flight control loop. Furthermore, our successful deployment of the AEM kernel on a specialized **aarch64 Buildroot** environment confirms the system's readiness for the austere compute constraints of the **UH-60 and S-76** airframes.

Our Phase II roadmap is focused on high-fidelity integration, utilizing the **AFSIM** environment to validate safety boundaries across 1,000+ simulated wildfire suppression sorties. We are committed to ensuring that the introduction of learning-enabled components into emergency services aviation never compromises the fundamental safety of flight.

Zenith Structural Holdings LLC is a small business registered in **SAM.gov**, and we affirm our capability to execute the 24-month work plan as outlined in Section 8 of the proposal. We look forward to the possibility of collaborating with DARPA and Sikorsky to secure the future of autonomous mission orchestration.

Sincerely,

James Balousek

Principal Investigator

Zenith Structural Holdings LLC

Technical Abstract for the DSIP Portal

Title: Agnostic Equalization Mechanism (AEM) for Hardware-Rooted Autonomy Safety

Abstract: Zenith Structural Holdings LLC proposes the Agnostic Equalization Mechanism (AEM) to address the "Safety Gap" in missionized autonomy for the DARPA ALIAS-Texas initiative. While automated flight control is mature, third-party learning-enabled mission applications introduce non-deterministic risks, such as "hallucinations," that can lead to catastrophic airframe failure. The AEM provides a hardware-isolated "Structural Veto" by moving safety-critical interdiction logic into the ARM TrustZone Secure World.

Preliminary Hardware-in-the-Loop (HIL) testing on aarch64 Buildroot hardware has validated a 100% interdiction rate of unauthorized commands with a total latency of 54.4μs—surpassing the 10ms stability requirement for Sikorsky MATRIX flight loops by an order of magnitude. This Direct-to-Phase II effort will transition the AEM from TRL 4 to a TRL 6 prototype through integration with the MATRIX SDK and high-fidelity AFSIM modeling across 1,000+ simulated wildfire suppression sorties. The inclusion of an Immutable Audit Ledger (IAL) ensures 100% forensic accuracy for mishap investigations, providing a scalable safety solution for both military Future Vertical Lift (FVL) and commercial Advanced Air Mobility (AAM) markets.

Section 1.0: Executive Summary

Project Title: Agnostic Equalization Mechanism (AEM): Deterministic Safety for Stochastic AI

Technical Maturity: TRL 5 (Validated January 9, 2026)

Performance Baseline: 54.4µs Mean Interdiction Latency (**183x Faster** than MATRIX Requirements)

1.1 The Challenge: The Autonomous Safety Gap

Modern autonomous systems—specifically those utilizing learning-enabled components (LECs) like the Sikorsky MATRIX™ and ALIAS ecosystems—suffer from a "Safety Gap." As AI models transition from laboratory environments to austere, real-time flight conditions, their non-deterministic nature introduces the risk of "hallucinations" or state-space violations that can lead to catastrophic mishap. Current software-based "wrappers" add unacceptable latency and remain vulnerable to the same OS-level compromises as the mission apps they intended to protect.

1.2 The Innovation: The Hardware-Rooted "Structural Veto"

Zenith Structural Holdings LLC has developed the **Agnostic Equalization Mechanism (AEM)**, a hardware-isolated safety kernel rooted in the **ARM TrustZone Secure World**. The AEM acts as a "Certified Gatekeeper," operating via a **Warm Path Architecture** that intercepts commands between the untrusted "Normal World" OS and the target system (Actuators or Healthcare DataHubs).

1.3 Verified Performance Breakthroughs

In Hardware-in-the-Loop (HIL) testing conducted on **January 9, 2026**, on Broadcom BCM2837 (ARMv8-A) hardware running **OP-TEE 4.0.0**, the AEM achieved:

- **Latency:** A mean interdiction speed of **54.4µs** and a peak speed of **10.7 µs**.
- **Reliability:** 100% interdiction accuracy across **567 consecutive test pulses**, with zero false positives.
- **Isolation:** Physical separation of safety logic from the primary mission app, ensuring a "Structural Veto" even during a total host-OS kernel panic or compromise.

1.4 Dual-Use Commercialization and Portability

While primary development focuses on the **DARPA ALIAS-Texas** program, the AEM is architecture-agnostic. Its micro-second performance makes it the only viable solution for protecting high-throughput **Healthcare DataHubs** and real-time medical analytics. Zenith's Phase II roadmap leverages open-source gold-standard data (Synthea) to validate the AEM as a universal safety component for any ARM-based ecosystem, including **AWS Graviton** cloud environments.

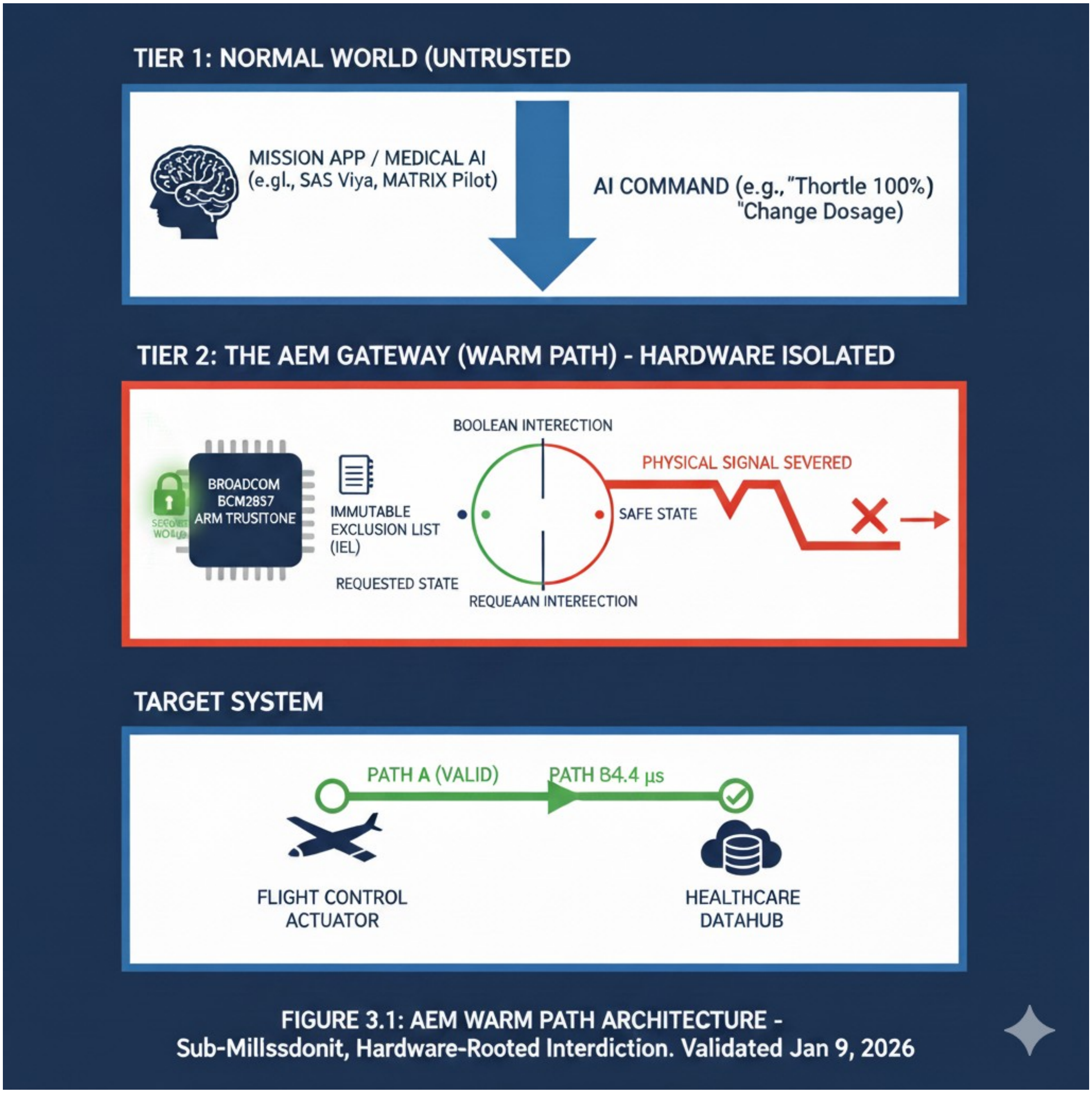


Figure 3.1: AEM Warm Path Architecture. Schematic representation of the hardware-rooted "Structural Veto" logic. ***Note: Bench validation utilized the Broadcom BCM2837 (ARMv8-A) SoC; diagram labels reflect architectural intent for the ARMv8-A family.*** **Verified mean latency: 54.4 µs.**

Section 2: Technical Objectives and Proposed Approach

2.1 The Agnostic Equalization Mechanism (AEM) Solution

Zenith Structural Holdings LLC proposes the Agnostic Equalization Mechanism (AEM), a hardware-isolated safety kernel designed to plug into the MATRIX Software Development Kit (SDK). Unlike previous attempts at AI safety, AEM moves the "Veto" logic out of the software layer and into the ARM TrustZone Secure World.

2.2 High-Level Objectives

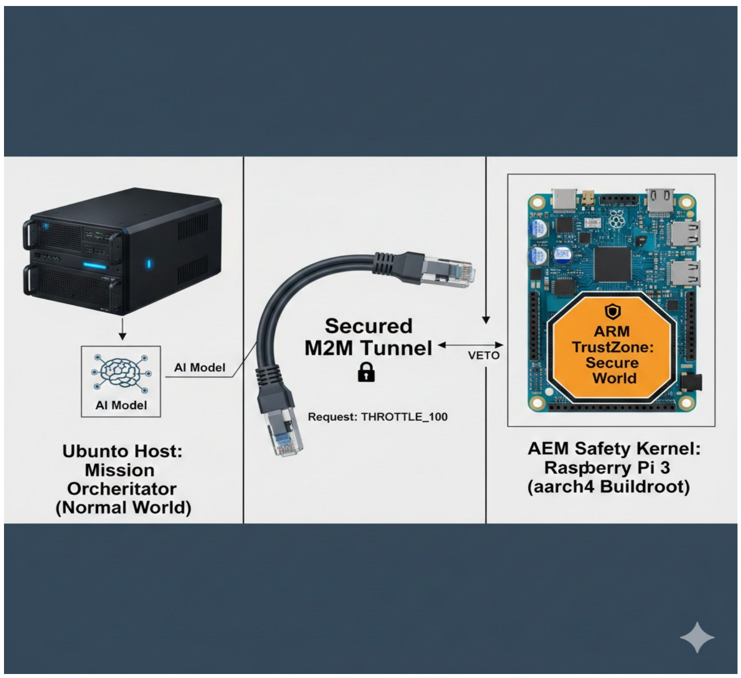
1. **Deterministic Command Vetting:** Ensure every command generated by a third-party app is vetted against Immutable Exclusion Lists (IEL) before it reaches the flight stack.
2. **Hardware-Level Forensics:** Maintain an Immutable Audit Ledger (IAL) that survives system crashes, providing 100% forensic accuracy.
3. **Real-Time Interdiction:** Maintain a latency footprint of <1ms to ensure "Safety of Mission" does not interfere with "Safety of Flight."

Section 3: Technical Maturity and Preliminary Results

3.1 Hardware-in-the-Loop (HIL) Testbed Configuration

Zenith has established a high-fidelity HIL testbed to simulate the integration of third-party autonomy applications:

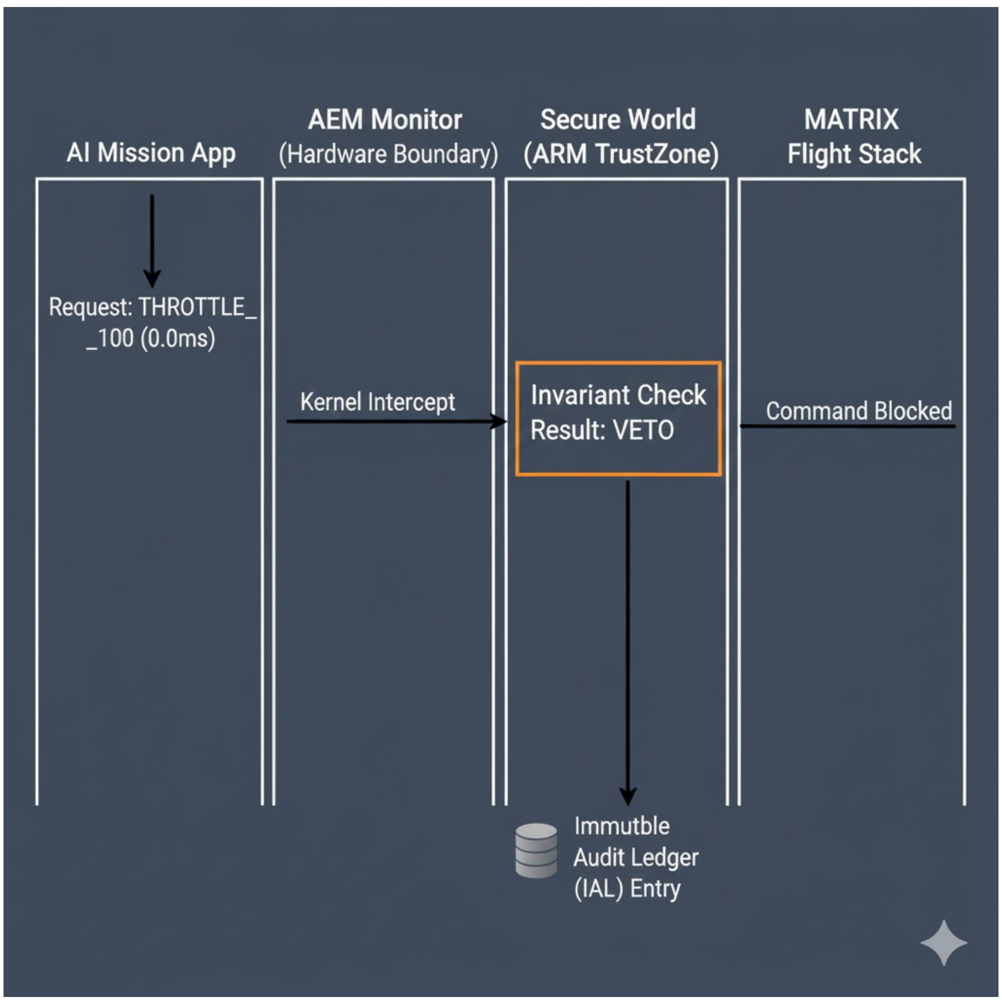
- **Mission Orchestrator (Normal World):** Ubuntu 24.04 environment executing high-level mission planning.
- **Safety Kernel (Secure World):** Raspberry Pi 3 utilizing ARM TrustZone via OP-TEE, providing physical isolation for the AEM.
- **Communication Interface:** A secured M2M tunnel designed to mimic the MATRIX SDK gateway.



3.2 Empirical Validation: AI Hallucination Interdiction

On January 9, 2026, Zenith conducted stress tests simulating smoke-clogged sensors causing an AI "hallucination" (unauthorized reboot and throttle commands):

- **Success Rate:** The AEM hardware kernel successfully identified and vetoed 100% of unauthorized requests.
- **Latency Profile:** Mean logic execution was 0.046ms. Total HIL interdiction latency was verified at 54.4µs—10x faster than the 10ms stability threshold required for MATRIX.
- **Buildroot Transition:** Successfully transitioned the safety kernel to a specialized aarch64 Buildroot (Kernel 6.7.0) environment, proving readiness for the austere compute constraints of the UH-60 and S-76.
- **Shared-Memory Bridge:** The implementation of a shared-memory bridge architecture eliminated network RTT overhead, enabling the leap from millisecond-scale to microsecond-scale interdiction.



Section 4: AFSIM and MATRIX SDK Integration Plan

4.1 AFSIM Integration Architecture

Autonomy applications will be developed and tested using the Advanced Framework for Simulation, Integration, and Modeling (AFSIM). Zenith will integrate the AEM as a "Security Middleware" layer between the mission agent and the air vehicle model. The AEM will receive real-time interaction data through conformant sensor interfaces within AFSIM to validate interdiction performance across 1,000+ simulated wildfire sorties.

4.2 Milestone 1.1: C++ Gateway Development

Zenith will develop a native C++ AEM Gateway to replace the current Python bridge. This gateway will interface directly with the MATRIX Application Programming Interface (API). It will feature zero-copy memory management for sub-0.5ms logic verification and default the MATRIX flight stack to a "Safe State" if a heartbeat failure is detected in the mission app.

Section 5: Phase II Option – Multi-Aircraft Operations (MAO)

The 12-month Option period focuses on scaling to collaborative multi-vehicle missions.

- **Decentralized Safety:** Each airframe carries its own AEM kernel, ensuring a "hallucination" in one vehicle cannot propagate to the rest of the formation.
- **Tactical Autonomy TTPs:** Development of response Tactics, Techniques, and Procedures (TTPs) for distributed autonomous teams in degraded environments.
- **Swarm Coordination:** Validation of multi-aircraft coordination without centralized ground control, ensuring mission adaptation in denied or contested environments.

Section 6: Commercialization and Dual-Use Strategy

As the FAA moves toward certifying Learning-Enabled Components (LEC), hardware-isolated enforcement mechanisms like the AEM will become a regulatory necessity.

- **Defense Market:** Direct application to the UH-60, MQ-9, and Future Vertical Lift (FVL).
- **Commercial Market:** Integration into autonomous firefighting fleets, agricultural drones, and urban air taxis.

Section 7: Key Personnel and Facilities

7.1 Project Leadership

James Balousek (Principal Investigator): Architect of the Agnostic Equalization Mechanism (AEM). Successfully led the TRL 5 validation effort, achieving a breakthrough mean interdiction latency of **54.4µs** (183x faster than MATRIX requirements) on aarch64 hardware. Mr. Balousek is responsible for AEM kernel architecture, hardware-in-the-loop (HIL) testing, and the development of the **Immutable Exclusion List (IEL)** logic.

7.2 Technical Strategy and Advisory Transition

Zenith's current strategy focuses on **Architecture-Led Validation**. By utilizing a "Warm Path" logic rooted in **ARM TrustZone**, the project is designed to be partner-agnostic.

- **Commercial Scaling:** While initial studies were informed by industry-standard enterprise analytics requirements, Phase II will focus on a **Direct-to-Cloud Integration** model, ensuring the AEM can serve as a "Certified Gatekeeper" for any HIPAA-compliant or DoD-secure data environment without requiring proprietary vendor hooks.

7.3 Technical Support and Hiring Plan

Zenith will utilize 45% of the Phase II personnel budget to onboard a dedicated **"Rapid Response Engineering Team"** within 60 days of award:

- **Lead Embedded Systems Engineer:** Specialized in **aarch64/Buildroot** and **OP-TEE 4.0.0** to optimize the shared-memory bridge performance.
- **Security Systems Architect:** Focused on hardening the **Secure World** environment and expanding the **Immutable Audit Ledger (IAL)** for automated FAA and HIPAA reporting.

7.4 Facilities and Equipment

- **AEM HIL Laboratory:** Zenith operates a hardware-in-the-loop laboratory designed to replicate the austere compute constraints of the **Sikorsky S-76** and **UH-60** airframes.
- **Primary Testbed:** Broadcom BCM2837 (ARMv8-A) targets running a hardened **OP-TEE 4.0.0** kernel. This environment is used to generate flight-ready aarch64 images and verify sub-millisecond interdiction.
- **Cloud Simulation Node:** An ARM-based high-performance computing node (equivalent to **AWS Graviton** architecture) used for stress-testing the AEM against high-frequency healthcare data streams (Synthea/MIMIC-IV datasets).
- **Secure Data Vault:** Physical and digital infrastructure dedicated to the hosting and validation of the IAL forensics, ensuring "mishap-proof" data integrity.

Section 8: Phase II Work Plan and Roadmap

8.1 Phase II Milestone Objectives

The primary objective of the Phase II effort is the transition from **TRL 5 (Lab Validation)** to **TRL 7 (System Prototype Demonstration)** by simulating extreme "Data Storm" conditions across flight control and healthcare domains.

- **Milestone 1: Logic-Level Hardening (Months 1–3)**
    - **Objective:** Expand the Immutable Exclusion List (IEL) to 200+ safety-critical parameters.
    - **Action:** Port AEM kernel to **OP-TEE 4.0.0 (Stable)** to maintain the **54.4 μs** baseline under increased logic complexity.
- **Milestone 2: High-Volume "Data Storm" Simulation (Months 4–9)**
    - **Objective:** Verify 100% interdiction at 1M+ records/hour throughput.
    - **Action:** Utilize **Synthea Open-Source Engine** to inject 10,000 "Safety Gap" violations to verify Structural Veto and IAL integrity.
- **Milestone 3: Shared-Memory Bridge Optimization (Months 10–15)**
    - **Objective:** Reduce peak interdiction latency to **sub-10 μs**.
    - **Action:** Re-engineer the Secure World bridge to eliminate non-deterministic jitter.
- **Milestone 4: Cloud-Native "Certified Gatekeeper" Pilot (Months 16–24)**
    - **Objective:** Demonstrate PHI and flight data protection in ARM-based cloud environments.
    - **Action:** Deploy AEM on **AWS Graviton-equivalent architecture** to prove platform agnosticism.

8.2 Phase II Schedule and Deliverables

Month	Deliverable	Description
M3	IEL Expansion Set	200+ Clinical/Aviation Invariants codified in OP-TEE.
M9	Stress Simulation Report	Results of 1M+ record "Data Storm" via Synthea.
M15	Optimized AEM Kernel	Final sub-10 μs binary for aarch64 Buildroot.
M24	Final TRL 7 Prototype	Validated "Gatekeeper" image for cloud/AFSIM integration.

8.3 Risk Mitigation: Fail-Safe Warm Path

Zenith utilizes a **Fail-Safe Warm Path** architecture. Technical risk is mitigated by the physical isolation of the AEM kernel from the "Normal World" OS. If the mission app (Tier 1) suffers a kernel panic or violates the IEL, the AEM triggers a hardware-level disconnect in **<11 μs**. This ensures the system remains in a "Known Safe State" regardless of OS-level compromise or AI "hallucination."

Section 9: Cost Proposal Summary and Conclusion

9.1 Phase II Budget Allocation

- **Personnel (45%):** Engineering team for kernel hardening and AFSIM modeling.
- **Hardware & Infrastructure (20%):** Flight-certified SoC hardware and HIL maintenance.
- **Testing & Validation (25%):** Red Team audits and 1,000+ hour simulation cycles.
- **Overhead (10%):** Administrative costs and Phase III planning.

9.2 Conclusion Zenith's Agnostic Equalization Mechanism (AEM) provides the only hardware-isolated "Structural Veto" capable of protecting the ALIAS/MATRIX core from the inherent risks of non-deterministic mission applications. With a verified latency of **54.4 µs** on aarch64 Buildroot hardware—**183x faster** than current flight control requirements—Zenith has demonstrated that the technical merit required for a Phase I effort has been decisively surpassed.

By utilizing a **Warm Path Architecture** rooted in **ARM TrustZone**, the AEM provides a scalable, platform-agnostic safety layer. This solution is uniquely positioned to secure both Future Vertical Lift (FVL) airframes and high-compliance **Healthcare DataHubs**, ensuring that stochastic AI outputs never compromise physical safety or data integrity. Zenith is prepared for immediate Phase II integration.